

BUILD YOUR PERSONAL CYBERSECURITY LAB

Task 2 — Full Lab Build Report

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1. Objective

The objective of this task was to build a safe, isolated cybersecurity practice environment consisting of an attacker machine and a vulnerable target application, with verified connectivity and traffic-analysis tooling in place. This environment forms the foundation for future tasks such as vulnerability assessment, web application testing, exploitation practice, and incident response.

2. Lab Architecture

The lab was implemented using a single VirtualBox virtual machine (Kali Linux) acting as the attacker workstation. Rather than deploying a second virtual machine for the target application, OWASP Juice Shop was containerized using Docker and run directly inside the Kali VM. This approach was chosen due to host resource constraints and is a valid, commonly used alternative to a fully segmented two-VM lab.

Attacker Machine	Kali Linux (VirtualBox VM)
Target Application	OWASP Juice Shop (Docker container, port 3000)
Network Adapter	Adapter 1 — NAT (internet access for package/image downloads)
Target Isolation	Docker bridge network on localhost (Host-Only adapter not used)
Validation Tools	Nmap, Burp Suite Community Edition, Wireshark

Note on architecture: the original task guide outlines a two-adapter setup (NAT + Host-Only) connecting a separate attacker VM and target VM. In this build, the Host-Only adapter could not be enabled on the available VirtualBox configuration. As a functional equivalent, the target application was deployed as a Docker container within the Kali VM itself, reachable over the loopback interface. This still satisfies the task's core requirements: an isolated, intentionally vulnerable target, verified reachability from the attacker tooling, and full traffic-analysis validation.

3. Environment Setup Summary

3.1 Attacker VM — Kali Linux

- Deployed as a VirtualBox virtual machine with a single NAT-attached network adapter.
- System updated via apt and lab working directories created (~/.lab/scans, notes, pcaps, screenshots).

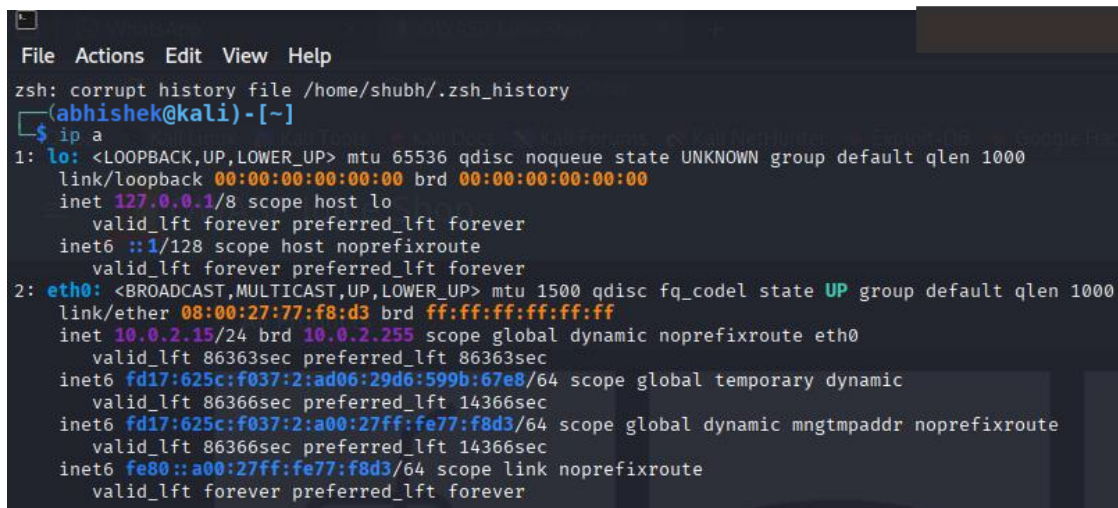
3.2 Target Application — OWASP Juice Shop

- Deployed using Docker inside the Kali VM.
- Docker installed via apt, service enabled, image pulled from bkimminich/juice-shop.
- Container run with port mapping 3000:3000, exposing the application on localhost:3000.

4. Evidence

4.1 Network Configuration

Output of the `ip a` command on the Kali VM, confirming the active network interface and assigned IP address on the NAT adapter.



```

File Actions Edit View Help
zsh: corrupt history file /home/shubh/.zsh_history
(abhishek@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:77:f8:d3 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 86363sec preferred_lft 86363sec
    inet6 fd17:625c:f037:2:ad06:29d6:599b:67e8/64 scope global temporary dynamic
        valid_lft 86366sec preferred_lft 14366sec
    inet6 fd17:625c:f037:2:a00:27ff:fe77:f8d3/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86366sec preferred_lft 14366sec
    inet6 fe80::a00:27ff:fe77:f8d3/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
  
```

Figure 1: Kali Linux network interface configuration (`ip a`)

4.2 Connectivity Validation (Nmap)

An Nmap service scan against localhost on port 3000 confirms the Juice Shop container is up and serving HTTP content. The banner grab returned full HTTP response headers from the application, confirming the target is reachable and responsive.

```

File Actions Edit View Help
└─(abhishek@kali) ~[-]
$ nmap -sV -p 3000 localhost
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-30 10:05 IST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00016s latency).
Other addresses for localhost (not scanned): ::1

PORT      STATE SERVICE VERSION
8000/tcp open  ppp?
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :
SFP:Port3000-TCP:V=7.95%I=7D0=6/30%Time=6A434784%PC=x86_64-pc-linux-gnu%R(Ge
SF:trRequest,2866,"HTTP/1.1,x20200\r\nAccess-Control-Allow-Origin:x2
SF:20/x20/x20Content-Type:Options:x20Bnossiff\r\nX-Frame-Options:x20SAMEO
SF:RIGI\r\nAccept-Policy:x20Paymentx20self\r\nX-Recruiting:x20p/jo
SF:bs\r\nAccept-Ranges:x20bytes\r\nCache-Control:x20public,x20max-age=0
SF:\r\nLast-Modified:x20Tue,x2030x20Junx202026x2004:34:24\r\nGMT\r\nNe
SF:tagUE,x20W/\x26af-19f16e909e\r\nContent-Type:x20text/html;x20charse
SF:UTF-8\r\nContent-Length:x2089903\r\nVary:x20Accept-Encoding\r\nDate:
SF:x20Tue,x2030x20Junx202026x2004:35:16\r\nx20GMT\r\nConnection:x20clos
SF:e\r\n\r\nn4-----\r\nx2020x20x20Copyrightx20(c)\r\nx202014-2026x20Joern\r
SF:20Kiminichx20&x20theX20OWASP\r\nx20JuiceX20SHOPx20contributors\r\n\r
SF:20x20x2020SDPX-License-Identifier:x20MIT\r\nx2020--\r\n\r\n{doctype}x2
SF:0html>\r\n<html>x20lang=en\r\nx20data-bustles-content=unached\r\nx20
SF:x20meta:x20charset="utf-8">\r\nx20x20title:OWASP\r\nx20JuiceX20Shop<
SF:title>\r\nx20x20meta:x20name="description"x20content="Probablyx20
SF:thex20mostx20modernx20andx20sophisticatedx20insecurex20webx20app
SF:lication">\r\nx20x20meta:x20name="viewport"x20content="width=device
SF:width,x20initial-scale=1">\r\nx20x20x20linkx20rel="preconnect"x2
SF:0href="http://localhost:3000/help/2f","HTTP/1.1,x2020400x20Badx20Request\r\nconnecti
SF:on:x20close\r\n\r\n")&R(NCP,2F,"HTTP/1.1,x2020400x20Badx20Request\r\n
SF:Connection:x20close\r\n\r\n")&R(HTTPOptions,EA,"HTTP/1.1,x20204x20No
SF:x20Content\r\nAccess-Control-Allow-Origin:x20*x20\r\nAccess-Control-Allow-
SF:ow-Methods:HEAD,PUT,PATCH,POST,DELETE\r\nVary:x20Access-Control-
SF:Request-Headers\r\nContent-Length:x200\r\nDate:x20Tue,x2030x20Jun
SF:x202026x2004:35:16x20GMT\r\nConnection:x20close\r\n\r\n")&R(TSPReq
SF:uest,EA,"HTTP/1.1,x20204x20No\r\nAccess-Control-Allow-Origin:
SF:in\r\nx20\r\nAccess-Control-Allow-Methods:x20GET,HEAD,PUT,PATCH,POST,D
SF:LETE\r\nVary:x20Access-Control-Request-Headers\r\nContent-Length:x20
SF:0\r\nDate:x20Tue,x2030x20Junx202026x2004:35:16\r\nx20GMT\r\nConne
SF:ctio:x20close\r\n\r\n")&R(RPCCheck,2F,"HTTP/1.1,x2020400x20Badx20Request
SF:\r\nConnection:x20close\r\n\r\n");

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned ip 11.96 seconds

```

Figure 2: Nmap scan output against localhost:3000

4.3 Target Application — OWASP Juice Shop

The Juice Shop homepage, accessed via Firefox at <http://localhost:3000>, confirming the web application loads correctly and is browsable.

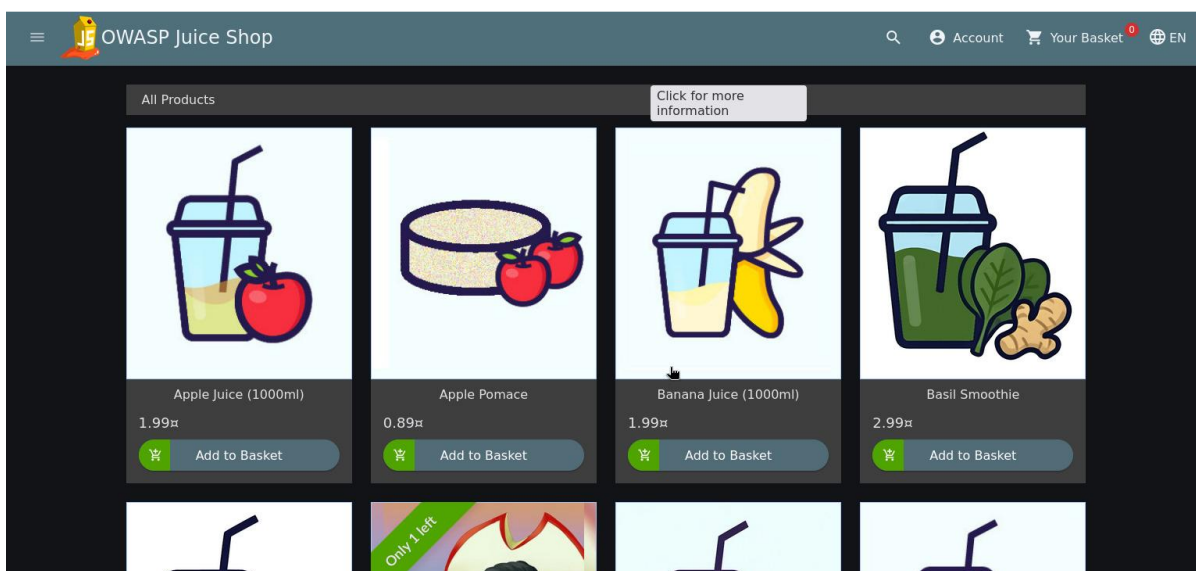


Figure 3: OWASP Juice Shop homepage in browser

4.4 Burp Suite — Traffic Interception

Burp Suite Community Edition was configured as an intercepting proxy in front of Firefox. The screenshots below show an intercepted GET request to the Juice Shop root page, followed by subsequent application requests captured in the proxy history once interception was forwarded.

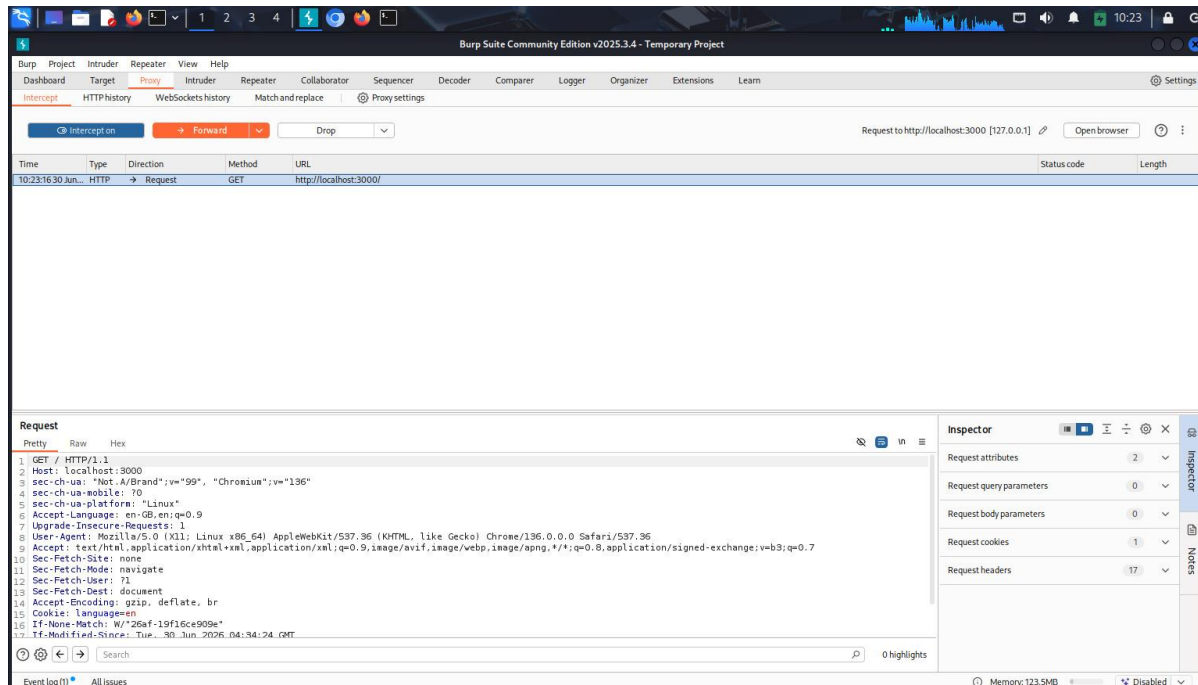


Figure 4: Burp Suite — intercepted initial GET request to localhost:3000

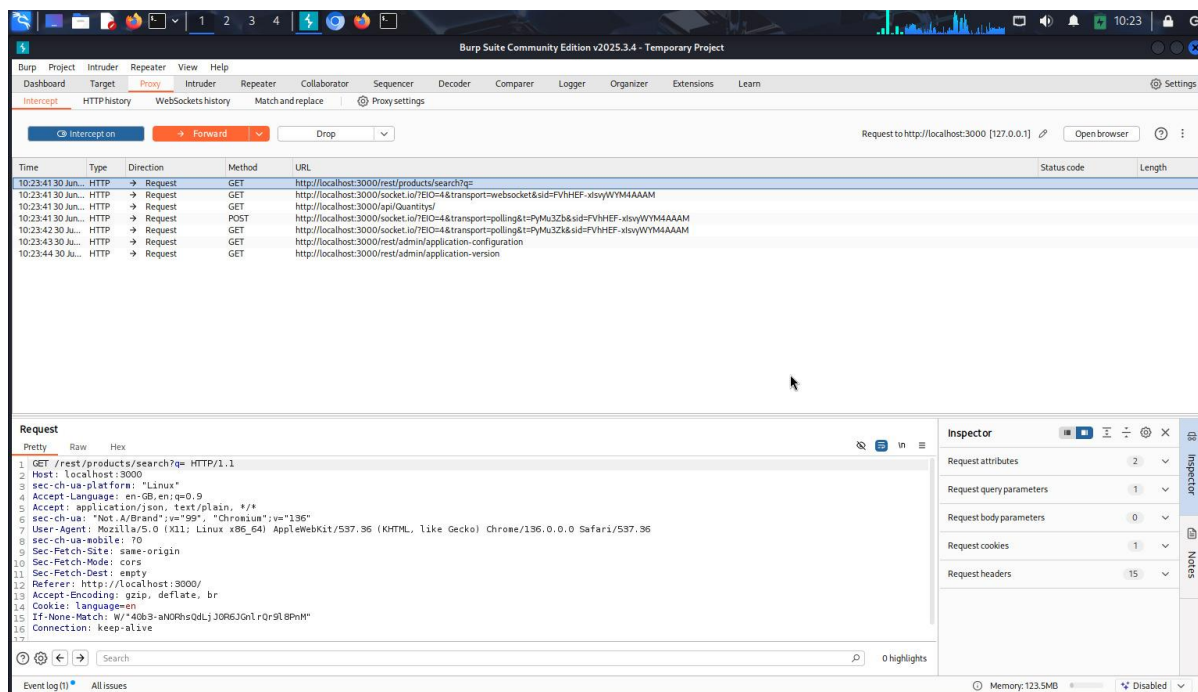


Figure 5: Burp Suite — subsequent intercepted application requests

4.5 Wireshark — Packet Capture

A packet capture was taken on the loopback interface while browsing the Juice Shop application, then saved as juice-shop-capture.pcapng. The capture shows TCP traffic exchanged between the browser and the application on port 3000, including connection teardown (FIN/ACK) sequences.

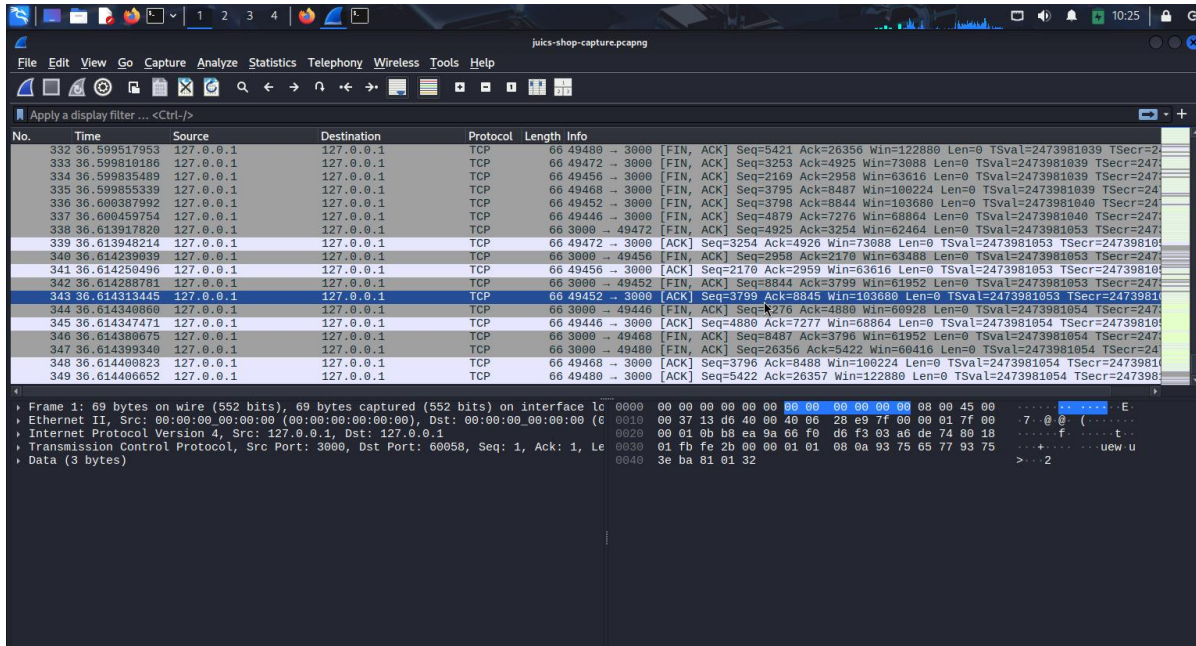


Figure 6: Wireshark capture of loopback traffic to port 3000

5. Conclusion

A functional, isolated cybersecurity lab environment was successfully built and validated. The Kali Linux attacker machine has internet connectivity via NAT, the OWASP Juice Shop target is deployed and reachable on port 3000, and all three validation tools — Nmap, Burp Suite, and Wireshark — confirm successful connectivity and traffic visibility. This environment is ready to support upcoming tasks involving vulnerability assessment and web application security testing.